PRE-FREEZE / DO NOT FABRICATE

CORE-2 rev.1 research schematic — ESP32-P4 + ESP32-C6. Do not order boards before P1 benchmark and spec/90 sign-off.
Project files: README.md, pinmap.md, power-tree.md, interfaces.md. GPIO conflicts: 0 (check_pinmap.py).

POWER: USB-C, BQ24074 charge/path, BMS 1S, MT3608 x3 (+12/+5/BL20), MP1584 3V3_MCU, AP2112 x2, MC34063 -12VMCU: ESP32-P4 module slot, ESP32-C6 Wi-Fi/BLE (SPI esp-hosted), USB-C HS, microSD SDIO

MIDI TRS-A: 6N137 IN / 74HCT125 OUT; GATE drivers 2N7000; LED level shifter

AUDIO: PCM5102A DAC + PCM1808 ADC (I2S), TL072 LINE/PHONES, rear jacks

CV: 2x MCP4728 (0x60/0x61), 6x TL074 x4.9, CV IN level shift, MCP3208 + chains

PANEL-2 + REAR I/O: J14 IDC-26 control, J15 FFC-40 LCD, J16 IDC-16 digital exp, rear jacks TB1/TB2

TEST POINTS (TP1..TP22)

POWER FLAGS (net drivers for ERC)

MODULE BUS: 2x IDC-26 analog bus (+-12V, CV1..4, GATE, audio), per-port protection

Boilerplate open hardware - CERN-OHL-S-2.0

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